QUICKSTEP COMPUTER CENTER

National Accreditation Board of Education Training.
(NABET)- Quality council of India) An ISO 9001:2008

MERN Stack Roadmap: Build Real-World Apps

Phase 1: Frontend Fundamentals (Days 1-30)

Week 1: Introduction to Web Development & HTML/CSS

- Course overview and MERN Stack basics
- HTML essentials (structure, tags, forms)
- CSS essentials (selectors, box model, Flexbox, grid)
- Responsive design principles

Week 2: JavaScript Basics

- JavaScript basics (variables, data types, operators)
- Control flow (loops, conditionals)
- Functions and scope
- DOM manipulation and event handling

Week 3: Advanced JavaScript & ES6 Features

- Arrow functions, template literals, Destructuring
- Array methods (map, filter, reduce)
- Object-oriented programming (OOP) basics
- Asynchronous JavaScript (callbacks, promises)

Week 4: Introduction to React

- Setting up React (Create React App)
- React components (class vs. functional)
- JSX and props
- State management and events

Phase 2: React Deep Dive & Frontend Integration (Days31-60)

Week 5: React Hooks & Advanced Components

- Introduction to React Hooks (useState, useEffect)
- Custom hooks
- Controlled vs. uncontrolled components
- Form handling in React

Week 6: React Router & Context API

- React Router (routing basics, nested routes)
- Dynamic routing and navigation
- State management with Context API
- Building a multi-page application

Week 7: Working with APIs & Axios

- Introduction to REST APIs
- Fetching data with Axios and handling errors
- CRUD operations in React
- Displaying and managing API data in components

Week 8: Project 1 – React Frontend Project

- Developing a small project (e.g., task manager, weather app)
- State management and routing
- Integrating an external API
- Styling and finishing touches
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Phase 3: Backend with Node.js and Express (Days61-75)

Week 9: Introduction to Node.js & Express
- Introduction to Node.js, npm, and Express
- Setting up a basic server
- Middleware functions in Express
- Handling HTTP requests and responses

Week 10: REST API Development
- Building a RESTful API
- CRUD operations and routing in Express
- Working with Postman for API testing
- Error handling and status codes

Week 11: MongoDB & Mongoose Basics
- Introduction to MongoDB and NoSQL databases
- Setting up MongoDB locally or with Atlas
- Defining schemas and models with Mongoose
- CRUD operations in MongoDB using Mongoose

Week 12: Authentication & Authorization
- Introduction to JSON Web Tokens (JWT)
- Implementing user authentication (login, signup)
- Protecting routes with JWT
- User roles and Authorization basics

Phase 4: Full Stack Integration & Deployment (Days76-90)

Week 13: Integrating Frontend and Backend
- Connecting React with the Node.js API
- Setting up proxy and handling CORS
- Sending and handling HTTP requests from React
- Managing frontend state with backend data

Week 14: Advanced Features & Security
- Data validation and sanitization
- Error handling and logging
- Implementing rate limiting and secure headers
- Setting up environment variables

Week 15: Project 2 – Full Stack Application
- Developing a full-stack project (e.g., social media app, e-commerce platform)
- End-to-end integration of frontend and backend
- CRUD operations, authentication, and deployment prep

Week 16: Deployment & Best Practices
- Deploying frontend on platforms like Netlify or Vercel
- Deploying backend on Heroku or Render
- Environment configuration for production
- Final project presentation and Q&A